

Iniziato	martedì, 15 giugno 2021, 12:02
Stato	Completato
Terminato	martedì, 15 giugno 2021, 21:33
Tempo impiegato	9 ore 30 min.
Valutazione	Non ancora valutato

Domanda **1**

Completo

Punteggio
max.: 1,00

Construct a deterministic TM of the kind you prefer, which decides the following language:

$$\mathcal{L} = \{w \in \{0, 1\}^* \mid \text{every } 1 \text{ occurring in } w \text{ is immediately followed by a } 0\}$$

Study the complexity of TM you have defined.

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(q_init, ▷) → (q_0, ▷, R)
(q_0, 0) → (q_0, 0, R)
(q_0, 1) → (q_1, 1, R)
(q_0, □) → (q_halt, □, S)
(q_1, 0) → (q_0, 0, R)
```

The above TM takes linear time in the input length, as it just scans the input string—from start to, at most, the very end—checking if every 1 it encounters is followed by a zero.

Domanda **2**

Completo

Punteggio
max.: 1,00

You are required to prove that the following function f is in **FP**. To do that, you can give a TMs or define some pseudocode. The function is one that, given two lists $L = L_1, \dots, L_n$ and $P = P_1, \dots, P_n$ of rational numbers returns their scalar products.

INPUT: two lists of rational numbers $L = L_1, \dots, L_n$ and $P = P_1, \dots, P_n$

OUTPUT: $L_1 * P_1 + \dots + L_n * P_n$

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dot_prod = 0
for i = 1 to n do
    dot_prod = dot_prod + L[i] * P[i]
return dot_prod

```

1. LENGTH OF INPUT ENCODING

First of all, we take n as the input length. Then, to encode the input, we go through the following steps:

1.1. Write rational numbers as pairs (N, D) , with N integer and D positive integer, then encode the sign of N (1 bit), the modulus of N ($\sim \lg|N|$ bits), D ($\sim \lg D$ bits), and the separators (2 bits). Then go from the encoding over $\{0, 1, \#\}$ to the one over $\{0, 1\}$ by converting 0 to 00, 1 to 01, and # to 10: such encoding has length $O(\lg|N| + \lg|D|)$, which does not depend on n .

1.2. Encode lists L and P by encoding their elements separated by #, then go from this encoding over $\{0, 1, \#\}$ to the one over $\{0, 1\}$ as described in step 1.1. Both $\llbracket L \rrbracket$ and $\llbracket P \rrbracket$ have length $O(n)$.

1.3. Encode the pair (L, P) as $\llbracket L \rrbracket \# \llbracket P \rrbracket$ and then, again, go from $\{0, 1, \#\}$ to $\{0, 1\}$. The final encoding length is therefore $l = \llbracket (L, P) \rrbracket = O(2n) = O(n)$, which is polynomial in n .

2. NUMBER OF INSTRUCTIONS

Since the for loop is executed n times, the number of instructions is $O(n) = O(l)$, so polynomial in the input encoding length.

3. EXECUTION TIME OF EACH INSTRUCTION

The only nontrivial instructions are multiplications between rational numbers in the input lists, which can be performed in polynomial time in l , and summations of such products, which, again, can be performed in polynomial time.

4. SIZE OF INTERMEDIATE RESULTS

i is a natural number $\leq n$, so it requires $O(\lg n) = O(n) = O(l)$ bits to be encoded, i.e. has polynomial encoding length in l . dot_prod is a summation of products of rational numbers in the input lists, so it has polynomial length too.

Domanda **3**

Completo

Punteggio
max.: 1,00

We studied the problem INDSET. You are required to classify the subset ONEINDSET of INDSET whose elements are pairs G, k such that G is an undirected graph such that every node is contained in *at most one edge*, and k is a natural number. To which complexity class(es) does ONEINDSET belong?

THEOREM. Let $G=(V,E)$ be an undirected graph. If every v in V is contained in at most one edge, then G has an independent set of cardinality $|V| - |E|$.

PROOF.

If $\{u,v\}$ is an edge, then we know by hypothesis that there is no other edge involving nodes u and v . So every node different from v can belong to the same independent set as u and every node different from u can belong to the same independent set as v . In other words, every edge induces two candidate independent sets of cardinality $|V| - 1$. If we repeat this process for all edges and apply it to each of those two subsets individually, we eventually get two independent sets of cardinality $|V| - |E|$.

The content of the above theorem allows us to devise a simple TM deciding ONEINDSET: the TM takes in input G,k and checks whether $|V| - |E| \geq k$ (INDSET requires the independent set to have cardinality **at least* k*): if so, returns 1, otherwise returns 0. Such TM trivially works in polynomial time, therefore ONEINDSET is in P.

◀ [Question Examples](#)

Vai a...